

Privacy Preserving in Data Mining

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Abstract

The protection of privacy in data mining had become an essential requirement for sharing confidential information for the purpose of data analysis, validation and publication. This essay provides a comprehensive overview and systematic interpretation of a compiled list of literature that examines the basic concepts of existing privacy-preserving data mining techniques, their strengths and weaknesses. The essay concludes with a discussion on the future of privacy-preserving data mining applications at an individual level.

1. Introduction

Over the past decade, data has become increasingly vital and now plays a crucial role in the creation of knowledge, the conduct of business and the implementation of governance. Data serves as the primary input to the systems used by individuals, institutions, businesses and science and gain an understanding by explaining, managing, regulating and predicting the world around us (Kitchin R et al., 2014). On an everyday basis, a massive amount of data is being sourced and collected from the public to able to derive knowledge about them. Mining datasets can potentially form patterns, and clusters or can be based on classification. However, during the data mining process, sensitive information belonging to individuals is often disclosed to multiple parties including data collectors, owners, users and miners (Ratra R et al., 2020).

The information that belongs to an individual or personal data can be identified or identifiable. This concept encompasses the aspects that data related to a person and that this person can be identified. One of the ways privacy can be compromised is when data mining techniques use non-sensitive identifiers to link personal identifiers such as physical, physiological, mental, economic, cultural and other sensitive personal information (Ozkoc E, 2021).

Nevertheless, individuals do not feel disturbed or violated of their privacy when their data is utilized in a study that is perceived to be well-intentioned (Clifton C et al., 2002). What is overlooked in this context is the challenge that once information falls into the hands of an unauthorized party, it becomes practically impossible to prevent misuse. The concerns over privacy and data security in data mining arise as the data owners are reluctant to share data for analysis purposes which created a barrier in the data mining task.

Given the absence of a definitive authority to which one can turn for guidance, it is crucial that those involved in the creation, supply and use of data actively engage with ethical considerations. However, the development of privacy-preserving techniques offers a new direction for addressing the issue where the primary means of utilizing the data without compromising is to remove an individual's identifiability (Bhagat B et al., 2015).

The essay concentrates on privacy protection algorithms and challenges that arise with data mining issues. Further, exploring a potential solution for the existing issues.

2. Data Privacy

A considerable amount of literature has been published on privacy preservation in data mining algorithms and their methodologies. These studies state the efficiency of the proposed metrics for Privacy Preserving Data Mining (PPDM) as it examines the privacy levels and data quality based on specific aspects of privacy.

Before delving into an explanation of privacy-preserving data mining methods, it is essential to understand the data that is being handled. Data controllers store and use the data collected through digital applications using appropriate methods and when necessary share them after anonymising the data. Pramanik M et al. (2021) categorized the collected data into four groups:

Identifiers (ID): This group includes information that directly and uniquely identifies individuals, such as their full name and social security number.

Quasi-identifiers (QID): These are identifiers that, when combined with external data, enable the indirect identification of an individual. Non-unique attributes like gender, age and postal code are included in this group.

Sensitive attributes (SA): This group contains private and sensitive data pertaining to individuals such as medical history and salary information.

Insensitive attributes: This group includes general and non-risky data that do not fall under any other categories.

2.1 Methods of privacy-preserving

Privacy-Preserving Data Mining (PPDM) is the practice of conducting data mining on data sets sourced from databases that contain confidential and sensitive information in a multi-party setting well ensuring that the data of each party remains undisclosed to the other parties involved (Lindell Y et al., 2000). Likewise, Qi X et al. (2011) assert that privacy-preserving data classification can be based on data distribution, data distortion, data mining algorithms, rules hidden and privacy protection. He mentions that to achieve deriving high-quality data that protect privacy must be modified. This can be done through several methods such as

using adaptive heuristics techniques to modify only selected values of data, minimising information loss. Encryption technologies like securing multiparty computation can also be employed to ensure that each site only has knowledge of its input and not others, making calculations secure. Data reconstruction methods can reconstruct the original data distribution from random data to maintain privacy while preserving utility.

Research Direction	Demonstration
General privacy preservation technology	Perturbation, Randomisation, Swapping, Encryption
Data mining privacy preservation technology	Association Rule, Mining, Classification, Clustering
Privacy protection data publishing principle	k-anonymity, l-diversity, m-Invariance, t-Closeness

2.2 Data Perturbation

In preserving the statistical integrity, data perturbation uses randomization and micro aggregation methods to the original data. In their analysis of the randomization method, Huang Z et al. (2005) identified the impact of correlation on the privacy of a dataset that is disguised through random perturbation. They proposed two methods for reconstructing original data from a disguised dataset: one based on Principal Component Analysis (PCA) and the other on Bayes estimate (BE). By using PCA, they provided both an intuitive and theoretical explanation of how data privacy can be affected by correlation in a randomised data set. The study found that as the correlation of the data increases both the PCA-based and BE-based methods can reconstruct more accurate data. However, the BE-based method was consistently found to be superior to the PCA-based method. To combat data reconstruction techniques that exploit data correlation, a modified random perturbation method was proposed in which the random noises are correlated, The experiments revealed that the more the correlation of the noises resembled that of the original data the better privacy preservation can be achieved.

2.3 Anonymisation

Anonymization is a technique used to safeguard personal or private information by eliminating identifiable details from the original data. Various methods can be employed to perform data anonymization, with k-anonymization being a commonly used approach. This method involves ensuring that each row in a table cannot be distinguished from at least k-1 other rows based solely on a set of attributes. When this condition is met, the table is

considered to be k -anonymized with respect to these attributes. Previously, techniques to achieve p -sensitive k -anonymity relied on suppression and generalization. Additionally, this article introduced by Solanas A et al. (2008) proposes a new method of achieving p -sensitive k -anonymity using micro-aggregation. The paper introduces two heuristics to accomplish this goal, which overcome the shortcomings of generalization and suppression while maintaining a low level of information loss. Other sophisticated techniques such as l -diversity, (α, k) -anonymity, t -closeness, and m -confidentiality have been proposed to protect against attribute disclosure, but they all rely on generalization. Thus, the micro-aggregation approach presented in this paper would be a novel technique for achieving privacy preservation in all of these approaches.

2.4 Differential privacy

Furthermore, Dwork C (2008) proposed the differential privacy approach to protect the privacy of database query results. This model is designed to address attacks that can take place between the time a database query is sent and when a response is provided. Specifically, it aims to prevent the disclosure of a single record's existence across multiple databases by ensuring that it is not possible to determine which database responded to a query that was issued in multiple databases. In the same vein, Sanchez D et al. (2013) analysed a method for anonymizing data that incorporates the advantages of individual ranking micro aggregation, which minimizes information loss and makes no assumptions about data usage, with the strong privacy guarantees of ϵ -differential privacy. The method is designed for data sets with numerical attributes, to which Laplacian noise is added to satisfy the differential privacy requirement. The authors demonstrate that their approach reduces the scale parameter of noise needed to achieve differential privacy, leading to the better utility of the anonymized data outputs.

Other researchers, however, who looked at PPDM techniques, have established the hybrid technique, Bhagat B et al., (2015) for example, combining two or more methods to provide better privacy protection. It combines the randomization and generalization methods, where the raw data is randomized first and then generalized. This technique has improved accuracy and can reconstruct the original data with no loss of information. Before that, Murat Kantarcioglu and Chris Clifton introduced a hybrid technique that combines noise addition and secure multiparty computation (SMC), a cryptographic technique to securely association rules over horizontally partitioned data. They added a little noise in the form of false rules

while sharing encrypted rule sets with other parties. Additionally, the AES encryption technique was combined with anonymization to provide a higher level of security.

3. Shortcomings in the privacy-preserving methods

There are various techniques available to protect privacy in data mining, such as data distribution, distortion, anonymization, and more. However, these approaches still suffer from major shortcomings, including issues with data distribution to multiple providers, privacy in cloud computing, the integrity of mining results, data utility, scalability, and performance. Therefore, an efficient and scalable model is needed to overcome these problems, along with proper anonymization techniques that remove personal identification while still satisfying privacy requirements and preserving data utility. While K-anonymity has been an effective method, it has been shown to be vulnerable to certain attacks. Thus, future privacy-preserving data mining techniques must advance and combine different functionalities to satisfy the requirements of various clients and communities. Additionally, search algorithms should be scalable and efficient, able to handle large volumes of data, and able to preserve privacy and data integrity while still being compatible and useful (Aldeen Y et al. 2015).

4. Conclusion

As digital data continues to grow at an unprecedented rate, current privacy preservation methods are handling and struggling to keep up with the vast amounts of data. The field of privacy-preserving big data analytics is at the evolving stages, and there are concerns that these methods may not be able to guarantee user privacy due to issues related to operations and efficiency. To address this, this essay critically reviews various privacy-preserving methodologies used in big data analytics, intending to provide a comprehensive and systematic study of their techniques. By doing so, a better understanding of the state-of-the-art privacy-preserving methods and develop a roadmap for designing the next generation of privacy-preserving big data.

5. Reference

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